Zewail City of Science, Technology and Innovation

University of Science and Technology

School of Computational Sciences and Artificial Intelligence

CSAI 203 – Fall 2026 | Introduction to Software Engineering

Academic Course & Schedule Planner

Software Requirements Specification (SRS)

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1. Introduction

1.1 Purpose

This document outlines the contributions made by Member 1 to the Academic Course & Schedule Planner web application, which supports semester timetables, prerequisite checks, and GPA estimation [1]. It covers FR-01–FR-05, Use Case Diagram 1, UC-01–UC-05, and NFR-01. The intended readers are the project team, the instructor, teaching assistants, developers, and testers.

1.2 Scope

The system enables students to view the courses offered in a semester, compare prerequisites against their course history, create proposed schedules, identify time conflicts within the same week, and estimate their GPA from past grades and credits. It also provides schedule save/load functions and administration tasks.

Boundaries: The system is a planning aid and is not an official registration system. It does not enrol students, change university records, or link to the university's student information system (SIS). Prerequisite checks and GPA estimates are based on the history recorded in the application. If data is missing or invalid, GR-1 applies.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
Academic Course & Schedule Planner (System)	Web-based system, accessible to students, for academic planning.
Student / Administrator	A student builds schedules and tracks GPA; an administrator maintains offerings, sections, capacity/status, and prerequisite rules.
Application Data	Stored courses, sections, prerequisite rules, completed-course records, grades, and saved plans. It is not official university data.
Completed Course Record	A student-entered record of a completed course, including grade and credit hours.
Course / Section	A course is defined by a code, name, and credit hours. A section is a single offering of a course in a semester.
Meeting Time	A weekly class meeting defined by day, start time, and end time.
Proposed Schedule	A set of selected sections for one semester. It is a planning artifact only, not an official registration.
Prerequisite	A course that must be completed before another course, based on the stored rules.
Time Conflict	Two meeting times on the same day overlap. Back-to-back meetings are not a conflict.
GPA / Grade Point	GPA is the credit-hour-weighted average of the grade-point values of the grades earned. Grade points are the values assigned to grades.
Credit Hours	The weight given to a course in the GPA calculation.

Term / Acronym	Definition
Completed Course	A course that has been completed and counts toward the GPA according to the project's eligibility criteria.
FR / NFR / UC	Functional requirement, non-functional requirement, and use case identifiers (for example FR-01, NFR-01, UC-01).

1.4 References

1. Dr. Rakha, M. S. Course Project: CSAI 203 – Introduction to Software Engineering. Project handout, Fall 2026. Zewail City.
2. Dr. Rakha, M. S. Lecture #5: Requirements Engineering. Introduction to Software Engineering (CSAI 203), lecture slides. Zewail City.
3. IEEE Recommended Practice for Software Requirements Specifications (IEEE 830-1998). IEEE, 1998.

1.5 Overview

Section 1 provides the purpose, scope, definitions, and references. Section 2 describes the overall environment, constraints, documentation, and assumptions. Section 3 contains the specific requirements: general rules and functional requirements (3.1), use cases (3.2), the conceptual class diagram (3.3), non-functional requirements (3.4), and external interface requirements (3.5). The appendices contain the data dictionary and the glossary.

2. Overall Description

2.1 Operating Environment

The Academic Course & Schedule Planner operates as a web application accessible through standard web browsers (such as Chrome, Firefox, Safari, and Edge). It runs on a cloud- or server-hosted Node.js environment backed by a relational database and communicates securely over TCP/IP networks through HTTPS APIs.

2.2 Design and Implementation Constraints

- Web Standards: Implementation must rely on HTML5, CSS3, and JavaScript to guarantee standard web browser compatibility.
- Data Access: Students have read-only permissions regarding official course offerings, section capacities, and prerequisite rules.
- Accessibility: UI components must align with baseline WCAG accessibility guidelines.

2.3 User Documentation

- Tooltips: In-app contextual hints guide users through schedule building, conflict warnings, and GPA estimations.
- User Guide: A standalone PDF manual detailing course browsing, timetable creation, history tracking, and plan exporting.

2.4 Assumptions and Dependencies

- Data Integrity: Master course data, section times, and prerequisite rules are assumed to be correctly maintained by administrators.
- Student Input Accuracy: Prerequisite FR-03Build Semester Schedulvaluations and GPA estimations depend on the accuracy and completeness of available academic information, including student-entered course history and any relevant data retrieved from university information systems.
- Availability: System performance relies on constant hosting server and database availability.

3. Specific Requirements

3.1 Functional Requirements

3.1.1 General Rules (GR-1 – GR-3)

The following general rules apply to the functional requirements in this section.

GR-1 Data Quality: The system never accepts missing or invalid data as valid, and no default value is substituted. Output messages explicitly name the affected item. Data-retrieval failures are reported as errors rather than shown as stale cached data.

GR-2 Notices: Where a result requires a notice, the notice is displayed with the result: a data notice for missing, unrecorded, or unavailable data; a planning notice stating that proposed schedules are planning artifacts and not registrations; and an estimate notice on calculated GPAs.

GR-3 Read-Only Operations: FR-01, FR-02, FR-04, FR-05, FR-06, FR-08, FR-09, and FR-10 are read-only operations and do not modify any course, section, prerequisite, history, or grade information. Only the active proposed-schedule workspace (via FR-03 and FR-07) and saved plans (via FR-11) are modified.

3.1.2 Functional Requirement Specifications

FR-01 — View Course Offerings

Description: The system shall enable a student to choose a semester and display the courses and course sections offered in that semester.

Use case: UC-01

Priority: High

Inputs: Semester selection (required); course code and/or course-name keyword filters (optional).

Processing: (1) Validate the selected semester. (2) Fetch the courses that have active sections in that semester. (3) Apply the partial filters, matching case-insensitively. (4) Sort the courses by course code, ascending.

Outputs: For each course: code, name, and credit hours. For each section: section ID, meeting days/times, and capacity/status when recorded (otherwise “not available”). A data notice (GR-2) or a “no matching courses” message.

Alternative / Exception Handling: An unselected or invalid semester prompts the user to select one. An invalid filter shows a validation message without changing the current view. An unrecorded semester displays a notice. A data-source failure displays an error with a retry option (GR-1).

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-01.1	Given semester S1 (courses A, B, C) and semester S2 (course D), selecting S1 shows A, B, and C and hides D.
AC-01.2	Each course shows its code, name, and credit hours; each section shows its meeting days/times plus capacity and status (or “not available”).
AC-01.3	Filtering S1 by code B returns only B. A query that matches nothing returns the “no matching courses” message.

ID	Acceptance Criterion / Expected Behavior
AC-01.4	Submitting without a semester selection shows a “select a semester” prompt. An invalid filter value shows a validation message and keeps the previous view.
AC-01.5	If no offerings are recorded for a semester, an explicit unrecorded notice is shown.
AC-01.6	A system or data-source failure shows an error message and removes stale cached data (GR-1).

Dependencies & Notes: Full-section representation (administrator status versus seat count) is subject to team agreement. The maximum filter length is defined in the design.

FR-02 — Verify Course Prerequisites

Description: The system shall return exactly one result stating whether the student’s recorded completed courses satisfy the selected course’s prerequisites: satisfied, not satisfied, or cannot be determined.

Use case: UC-02 (also invoked by FR-03)

Priority: High

Inputs: Student’s course selection, completed-course records, and course prerequisite rules.

Processing: (1) Retrieve the course prerequisite rules and the student’s history. (2) Check that the data is available; if not, return “cannot be determined”. (3) Check the prerequisites against the completed records. (4) Classify the status.

Outputs: Status (“satisfied”, “not satisfied”, or “cannot be determined”); per-prerequisite status list; list of unmet prerequisites (if “not satisfied”); data notice (GR-2) (if “cannot be determined”).

Alternative / Exception Handling: A course with no prerequisites reports “satisfied” (none required). Unentered or incomplete history, or an unentered or missing rule, reports “cannot be determined” (never falsely satisfied or unsatisfied). A data failure (GR-1) shows an error without a result.

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-02.1	Target X requires A and B: a student with records for A and B receives “satisfied”, with A and B listed as met.
AC-02.2	Target X requires A and B: a student with A but not B receives “not satisfied”, with B listed as unmet and A as met.
AC-02.3	A target course recorded with no prerequisites returns “satisfied” (none required).
AC-02.4	If the history is not entered and/or the prerequisite rule is not defined, the result is “cannot be determined”, never “satisfied” or “not satisfied”.
AC-02.5	Prerequisite evaluation is read-only and does not change any course, history, or rule records (GR-3).
AC-02.6	A data-source failure displays an error and outputs no result (GR-1).

Dependencies & Notes: Rule structures (all of / any of), the grade threshold for “completed”, and the distinction between an empty history and an unentered history are subject to team agreement. Depends on student authentication and completed-course history.

FR-03 — Create a Semester Schedule

Description: The system shall allow a student to create a draft semester schedule from available sections, and shall display warnings when prerequisites are not met (FR-02), when sections conflict with each other (FR-04), when a section is full, and when section information is incomplete.

Use cases: UC-03 (includes UC-02 and UC-04)

Priority: High

Inputs: The selected semester; sections added or removed by the student.

Processing: (1) Take only sections of the selected semester. (2) After each addition or removal, run FR-02 on the affected section's course and FR-04 on all selected sections. (3) Check section capacity/status. (4) Set the status to "No issues detected" only if all checks pass without warnings; otherwise set "Has warnings". (5) Save using FR-11.

Outputs: Workspace grid (courses, sections, meeting times); warnings (unmet/undetermined prerequisites, time conflict, incomplete check, full section, missing meeting data); planning notice and data notice (GR-2); validation status.

Alternative / Exception Handling: A schedule containing conflicts, unmet prerequisites, a full section, or incomplete data is never considered valid. However, a schedule containing warnings may be saved as a proposed plan with the status "Has warnings". FR-11 handles empty-plan requests and save failures, and does not display a successful save if the save fails.

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-03.1	When three non-overlapping, prerequisite-satisfied, open sections are selected and saved (FR-11), the status is "No issues detected".
AC-03.2	If a section overlaps an already selected section, a conflict warning is shown for both sections and the status changes to "Has warnings".
AC-03.3	Adding a section with an unmet prerequisite shows a prerequisite warning; adding a full section shows a full-section warning.
AC-03.4	A section with an unrecorded meeting time shows an incomplete-data warning, and the validation status is never "No issues detected".
AC-03.5	All schedule views display the planning notice, and building a schedule does not change any section or course-history data (GR-2, GR-3).
AC-03.6	When a conflicting section is removed, its warning disappears and the validation status is immediately re-evaluated.
AC-03.7	If no section is selected, attempting to save prompts the student to select sections and saves nothing.

Dependencies & Notes: FR-02, FR-04, FR-11, FR-12, and student identification. Team decisions are needed on whether plans with warnings can be saved, whether one or several plans per student/semester are retained, and how full sections are represented. The one-plan replacement rule must be consistent with FR-11.

FR-04 — Detect Schedule Conflicts

Description: The system shall list all pairs of sections and the days on which their meeting times overlap, together with the overlap intervals.

Use cases: UC-04 (also invoked by FR-03)

Priority: High

Inputs: Selected sections in the proposed-schedule workspace, each with weekly meeting times (day, start time, end time).

Processing: (1) Validate the meeting times (valid day, start earlier than end). (2) Compare the meeting times of all section pairs on the same day. (3) Conflict condition: same day AND start1 < end2 AND start2 < end1. Back-to-back meetings (end1 = start2) are not a conflict. (4) Compute the overlap interval [max(start1, start2), min(end1, end2)].

Outputs: Course codes, section IDs, day of conflict, and overlap interval. If there are no conflicts and all data is valid: “No conflicts detected”. If data is missing or invalid: “Incomplete check”, naming the affected sections.

Alternative / Exception Handling: Fewer than two sections reports “nothing to compare”. A missing or invalid meeting time is reported as an incomplete check (GR-1) and is never treated as conflict-free. A data-retrieval failure is shown as an error.

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-04.1	Sec A (Mon 09:00–10:30) and Sec B (Mon 10:00–11:30): reports a conflict between A and B on Monday with overlap 10:00–10:30.
AC-04.2	Sec A (Mon 09:00–10:30) and Sec B (Mon 10:30–12:00): reports no conflict (back-to-back meetings).
AC-04.3	Sec A (Mon 09:00–10:30) and Sec B (Tue 09:00–10:30): reports no conflict (different days).
AC-04.4	Sec A (Mon/Wed 09:00–10:00) and Sec B (Wed 09:30–10:30): reports a conflict on Wednesday only (09:30–10:00).
AC-04.5	Sec A (Mon 09:00–12:00) and Sec B (Mon 10:00–11:00): reports a conflict with overlap 10:00–11:00.
AC-04.6	With three sections where only A and B overlap, reports exactly the pair A and B.
AC-04.7	If a section has no end time, or its start time is greater than or equal to its end time, it is reported as incomplete and never considered conflict-free.

Dependencies & Notes: Pure algorithmic check; read-only. Called automatically by FR-03 whenever the timetable grid is modified.

FR-05 — Calculate GPA

Description: The system shall compute an estimated GPA as the credit-hour-weighted average of the grades of the completed courses recorded in the application. The result is an estimate and not an official GPA.

Use case: UC-05

Priority: Medium

Inputs: Student’s completed-course records (grade, credit hours); the project grade-point scale, eligible-course policy, and rounding rule.

Processing: (1) Identify eligible courses according to the policy. (2) Validate the records (credit hours must be > 0 and the grade must be on the scale). (3) Stop and report an error if any record is invalid

or incomplete. (4) If there are no eligible courses or the credit total is 0, show an explanatory message. (5) Compute the unrounded GPA with the weighted formula. (6) Round the displayed value.

GPA = (Σ g_i × c_i) / (Σ c_i), i = 1 … n

where g_i is the grade point and c_i the credit hours of course i, and n is the number of eligible courses. Expanded: GPA = (g1·c1 + g2·c2 + … + gn·cn) / (c1 + c2 + … + cn).

Outputs: Estimated GPA (rounded per the rule), courses included, credit hours, list of courses not eligible for inclusion, and the estimate notice (GR-2). If the GPA cannot be calculated: a clear error message stating the missing or invalid records.

Alternative / Exception Handling: Missing or invalid grade or credit hours: no GPA is displayed and the failing record is reported (GR-1). If there are no eligible courses or the total credits are 0, an explanatory message is shown. An undefined scale or rounding rule is reported as a system error.

Test Cases

Test	Included Courses (grade point g, credits c)	Unrounded Result	Calculation / Expected Behavior
T1	(g = 4.0, c = 3), (g = 3.0, c = 3), (g = 2.0, c = 2)	3.125	(12 + 9 + 4) / 8 = 25 / 8
T2	(g = 4.0, c = 3), (g = 0.0, c = 3)	2.0	12 / 6 = 2.0
T3	(g = 3.5, c = 4)	3.5	14 / 4 = 3.5
T4	(g = 3.0, c = 4), (g = 4.0, c = 2)	3.3333…	(12 + 8) / 6 = 20 / 6
T5	No eligible courses	No GPA	The message “No eligible courses” is displayed; no division by zero.
T6	Record with credit hours = 0, negative, or non-numeric	No GPA	Indicates an invalid record; no GPA is displayed.
T7	Record with a grade that is not on the GPA scale	No GPA	Indicates an invalid record; no GPA is displayed.

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-05.1	The displayed GPA matches the formula after applying the agreed rounding rule.
AC-05.2	Excluded courses do not count toward the result.
AC-05.3	GPA calculation is read-only (GR-3).
AC-05.4	The estimate notice (GR-2) appears with the output.

Dependencies & Notes: The grade scale, eligibility policy, rounding method (and number of decimal places), and letter-grade mappings are to be agreed by the team. For example, 3.125 rounds to 3.13 under half-up rounding and to 3.12 under half-even rounding.

FR-06 — View Course/Section Details

Description: The system shall let a student open a course or section from the FR-01 results and view its recorded details without changing the proposed schedule.

Use case: UC-06 (extends UC-01)

Priority: High

Inputs: Selected course or section identifier.

Processing: (1) Validate the selected identifier. (2) Retrieve the selected record. (3) Display the course code, name, credit hours, section identifier, meeting days and times, recorded capacity and status, and prerequisite information where applicable. (4) Mark any field that is not recorded as "not available"; it is not inferred.

Outputs: Details for the selected record; "not available" for any value not recorded.

Alternative / Exception Handling: An unknown or stale identifier shows an error and returns to the results. A retrieval failure shows an error and does not present cached details as current. Missing prerequisite data is never represented as "no prerequisites" (GR-1).

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-06.1	Selecting section S displays details belonging to S.
AC-06.2	Unrecorded meeting, capacity, or prerequisite fields are explicitly marked unavailable.
AC-06.3	Opening details does not modify the saved plan or the source records.

Dependencies & Notes: Extends FR-01 by presenting the detail view of the selected record. Uses GR-1 and GR-3.

FR-07 — Update Proposed Plan Selection

Description: The system shall support adding or removing a valid section in the student's proposed schedule and prevent the same section from being stored twice in that schedule.

Use case: UC-07 (extends UC-03)

Priority: High

Inputs: Selected semester; section identifier; add or remove action.

Processing: (1) Verify that the section belongs to the selected semester. (2) Reject a duplicate addition. (3) Apply the change. (4) Invoke the validation behavior defined by FR-03 (prerequisites, conflicts, section status, and data completeness).

Outputs: The updated proposed selection; refreshed warnings and validation status; a confirmation of the successful change.

Alternative / Exception Handling: A wrong-semester or unavailable section: the change is rejected and the system explains why. A duplicate addition: the section is not added again. Removal: warnings are recalculated. Incomplete data never produces "No issues detected" (GR-1).

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-07.1	A valid section is added once.
AC-07.2	Re-adding the same section creates no duplicate.
AC-07.3	Removing a section updates the plan and clears any warning caused solely by it.
AC-07.4	Official records and section capacity are unchanged.

Dependencies & Notes: Coordinates with FR-03 and does not define a second schedule-saving workflow. Whether a schedule with warnings may be saved remains pending team agreement (FR-03).

FR-08 — Review Plan and Credit-Hour Total

Description: The system shall display the selected sections in the proposed plan and calculate their total credit hours from valid recorded values.

Use case: UC-08 (extends UC-03)

Priority: High

Inputs: The current proposed schedule; the credit-hour values of the selected sections and courses.

Processing: (1) Retrieve the selected sections and their recorded credit-hour values. (2) Sum the credit-hour value of each distinct selected course according to the agreed course and section data model. (3) Do not treat missing or invalid values as zero, and do not double-count duplicate section records. (4) Recalculate after every addition or removal.

Outputs: The selected sections and their recorded meeting details; the total credit hours; a clear incomplete-data notice if any required value is missing or invalid.

Alternative / Exception Handling: Empty plan: the system displays zero selected sections and a zero total, if the agreed display policy permits. Missing or invalid credit hours: the system identifies the affected course and marks the total incomplete (GR-1). Retrieval failure: the system shows an error and no stale total as current.

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-08.1	For selected courses with 3, 3, and 4 credit hours, the system displays 10.
AC-08.2	Removing the 4-credit course changes the total to 6.
AC-08.3	Missing or invalid credit-hour data is reported, not silently counted as zero.

Dependencies & Notes: The credit-hour total is not a GPA; FR-05 remains responsible for GPA. Pending team agreement: whether credit hours are stored at course level or section level.

FR-09 — Review Conflict Results

Description: When FR-04 detects schedule conflicts, the system shall present each conflicting pair with the day and the overlapping interval in the proposed-plan view.

Use case: UC-09 (includes UC-04)

Priority: High

Inputs: The conflict-check output of FR-04 for the current proposed schedule.

Processing: (1) Obtain the latest conflict-check output of FR-04. (2) Use the FR-04 rule: same day AND start1 < end2 AND start2 < end1. (3) Preserve all reported pairs and each overlap interval. (4) Refresh the results after a section is added or removed.

Outputs: For each conflict: both course and section identifiers, the day, and the overlapping interval. If meeting data is missing or invalid: an incomplete-check warning naming the affected sections.

Alternative / Exception Handling: No conflict and all meeting data valid: the system shows the FR-04 no-conflict result. Fewer than two sections: the system shows that there is nothing to compare. The system never states that a schedule is conflict-free when the check is incomplete (GR-1).

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-09.1	Monday 09:00–10:30 and Monday 10:00–11:30 yield the overlap 10:00–10:30.
AC-09.2	Back-to-back or different-day meetings are not conflicts.
AC-09.3	Missing time data yields an incomplete status.
AC-09.4	Removing one of two conflicting sections refreshes the result.

Dependencies & Notes: A presentation and integration requirement for FR-04, not a new conflict algorithm. Pending team agreement: whether this requirement stays distinct or is merged into FR-04.

FR-10 — Review Validation Summary

Description: The system shall present a concise validation summary for the current proposed plan, consolidating the prerequisite, time-conflict, full-section, and incomplete-data findings produced by the existing validation workflow.

Use case: UC-10 (extends UC-03)

Priority: High

Inputs: The current proposed schedule; the validation results of FR-02, FR-03, and FR-04.

Processing: (1) After a selection change, collect the latest available validation results. (2) List each issue with the affected course or section. (3) Mark "No issues detected" only when all required checks completed with no warnings.

Outputs: The validation status ("No issues detected" or "Has warnings"); the issue list; the planning notice and data notice required by GR-2.

Alternative / Exception Handling: A check cannot be completed: the system identifies the affected item and shows a warning; unknown data is not treated as success (GR-1). Validation data cannot be retrieved: the system shows an error and does not show earlier results as current.

Acceptance Criteria

ID	Acceptance Criterion / Expected Behavior
AC-10.1	A valid plan with complete data and no issues displays "No issues detected".
AC-10.2	Any unmet prerequisite, conflict, full section, or incomplete data produces "Has warnings".
AC-10.3	Removing the only issue refreshes the status.
AC-10.4	The planning notice remains visible.

Dependencies & Notes: Coordinates with FR-03 and GR-1 to GR-3. It summarizes the existing validation and does not replace its rules.

FR-11 — Save Proposed Schedule

Description: Allows a student to store their active semester schedule in their user profile.

Inputs: Student identifier, selected semester, list of selected section IDs.

Processing: Verifies the active user session and a non-empty schedule grid, then writes the section IDs to storage.

Outputs: Save confirmation with timestamp.

Exceptions: If the workspace is empty, the save is cancelled and the user is prompted; an error message is shown if storage fails.

FR-12 — Load Saved Schedule

Description: Retrieves and displays a previously stored schedule for a chosen semester.

Inputs: Student identifier, chosen semester.

Processing: Fetches the saved section IDs, re-runs the conflict and prerequisite checks, and populates the grid view.

Outputs: Rendered schedule grid with current warnings.

Exceptions: Renders an empty grid if no saved plan exists; alerts the user if a saved section was removed from the offerings.

FR-13 — Clear Schedule Plan

Description: Empties all selected sections from the schedule workspace in a single action.

Inputs: Clear request, user confirmation.

Processing: Prompts the user to confirm; upon confirmation, clears the grid and resets the total credit hours to zero.

Outputs: Empty schedule workspace grid.

Exceptions: Cancelling the confirmation leaves the workspace unchanged.

FR-14 — Export Schedule Plan

Description: Generates a downloadable PDF document summarizing the current semester schedule.

Inputs: Active schedule sections and meeting times.

Processing: Formats section details, the weekly schedule, credit totals, and disclaimers into a PDF document stream.

Outputs: Downloadable PDF document file.

Exceptions: Export is disabled when the workspace is empty; an error message is shown if PDF generation fails.

FR-15 — Reset Student History

Description: Deletes all student-recorded completed courses and grades from the system.

Inputs: Student identifier, reset command, user confirmation.

Processing: Displays a confirmation dialog, deletes the course-history entries, and clears the GPA calculations.

Outputs: Success confirmation message and the reset history screen.

Exceptions: Data deletion is aborted if the user cancels the confirmation dialog.

FR-16 — Degree Progress Tracking & Degree Audit

Priority**:** Essential

When a student opens the Degree Progress page, the system fetches the student's requirements, UIS grades, and current position in the program (or an existing saved draft schedule, if one exists). It marks each course as complete, in progress, or planned; sums the credits; assigns each course to the Core, General Education, or Elective requirements; and computes the remaining credits, the percentage complete, and any courses that have become available through the FR-02 prerequisite check. The audit can be exported to PDF. When no program is found for the student, the advisor's contact information is displayed instead of an audit. When the UIS is unavailable or its data is stale (older than 24 hours), the age of the data is shown and the calculation uses the data that is available.

FR-17 — Advisor Override & Prerequisite Waiver Workflow

Priority: High

A student selects a section, chooses the override type (prerequisite waiver, credit-overload waiver, or late registration), and writes a justification. The system validates the request, creates it with the status PENDING, and sends it to the assigned advisor, who approves it, declines it, or requests additional information, recording the decision in a comment field. An approved override applies only to that student and section; the decision is audited and the student is notified (UC-18). Requests without a justification or for an unavailable section are refused, a second request for the same section is blocked, and a request stays pending while the advisor is unavailable.

FR-18 — Notification / Alert Management

Priority: High

When a business event occurs (for example, a seat becomes available from the waitlist, an override is decided, a registration window opens, or the schedule changes), the system selects the template and recipient, determines the recipient's channel preferences, creates an in-app notification and/or sends an email, and stores the delivery status and time. Failed email deliveries are retried according to the configured policy, and the notification is set to FAILED when retries are exhausted. When the user has disabled email, only the in-app notification is stored. An invalid recipient is recorded as a failure, and duplicate notifications for the same event key are not sent.

FR-19 — Role-Based Access Control & Permission Management

Priority: Essential

Every protected action is checked against the role's permissions (per action, with scope own, advisees, department, or all) and is denied by default. Unauthenticated requests receive HTTP 401, and authenticated but unauthorized requests receive HTTP 403. The System Administrator manages the role–permission mappings: invalid mappings are rejected, and every mapping change and every denied access is recorded in the audit log.

FR-20 — Academic Term & Registration Window Management

Priority: Essential

An authorized administrator sets up a term, including its start and end dates and the add/drop deadline, and defines registration windows with open and close times and the earned-credit ranges of the student groups they apply to. The system validates the date order and window overlaps, opens and closes windows automatically by server time, and checks each student's earned credits against the window range at registration. A reason is given when dates or ranges are invalid; a student outside the window or range is blocked with an explanation; a scheduling failure is logged and fails safely (registration closed).

3.2 Use Case Model

3.2.1 Use Case Diagrams

Figure 1 shows Use Case Diagram 1. The Student actor interacts with five core use cases of the Academic Course & Schedule Planner. “Check Course Prerequisites” and “Detect Schedule Conflicts” are included in “Build Semester Schedule”.

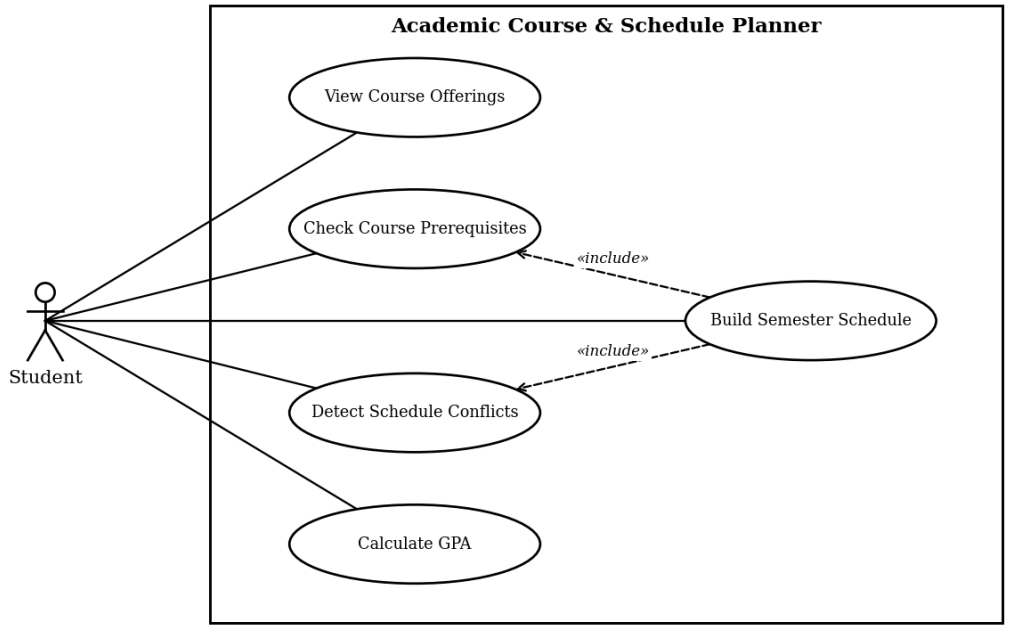


Figure 1: Use Case Diagram 1 — Student Academic Planning

Figure 2 shows Use Case Diagram 2 and Figure 3 shows Use Case Diagram 3.

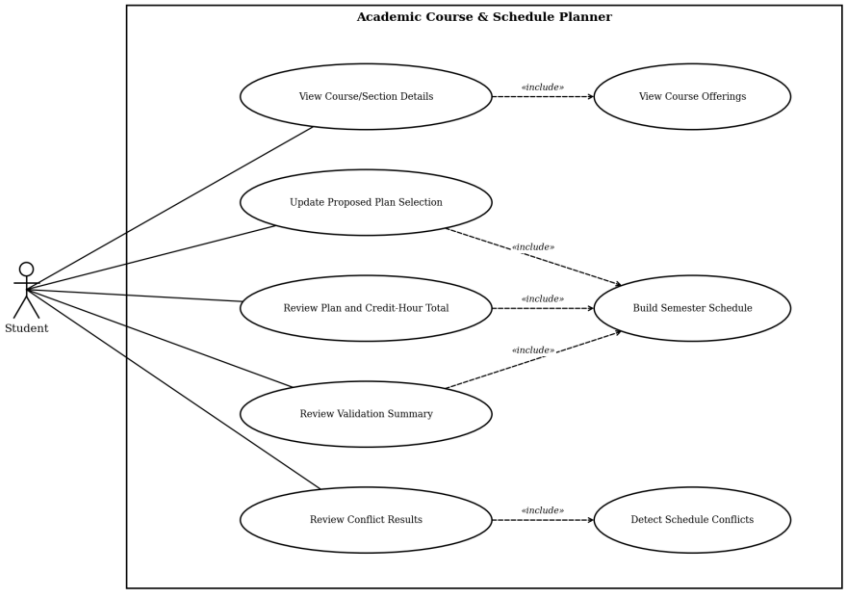


Figure 2: Use Case Diagram 2 — Plan Review

Use Case Diagram 2 shows the actor Student and the use cases UC-06 to UC-10 within the system boundary "Academic Course & Schedule Planner". It also shows View Course Offerings (UC-01), Build Semester Schedule (UC-03), and Detect Schedule Conflicts (UC-04), which remain the authoritative use cases for browsing, building, and checking the plan; the diagram refers to them and does not redefine them. View Course/Section Details extends View Course Offerings. Update Proposed Plan Selection, Review Plan and Credit-Hour Total, and Review Validation Summary extend Build Semester Schedule. Review Conflict Results includes Detect Schedule Conflicts. A proposed plan is a plan only and not an official university registration.

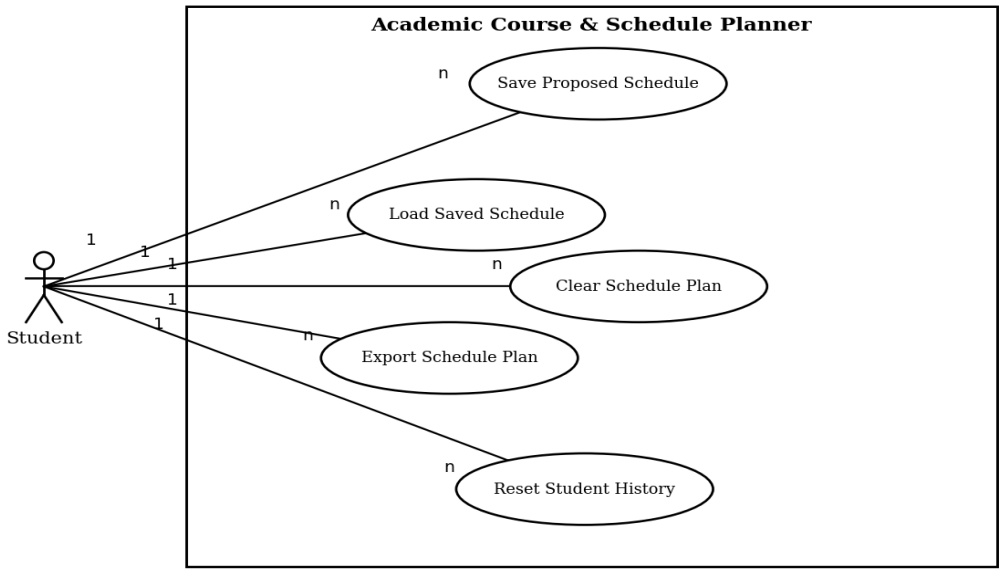


Figure 3: Use Case Diagram 3 — Schedule and History Management

3.2.2 Use Case Descriptions

UC-01 — Display Course Offerings

Actor: Student

Related Requirement: FR-01

Description: Search for the courses and sections in a chosen semester, optionally by code or name keyword.

Preconditions: The application is available.

Trigger: The student opens the course offerings view.

Main Success Flow:

1. The system displays the available semesters.
2. The student chooses a semester.
3. The system retrieves the recorded offerings.
4. Courses are listed by code, with their sections and any data notice, ordered by code.
5. The student optionally applies a code/name filter.
6. The system filters and displays the results.

Alternative Flows:

A1 (step 4): No offerings recorded → the system displays a notice and lets the student select another semester.

A2 (step 6): No matching courses → the system shows “no matching courses” and lets the student change the filter.

A3 (step 6): The student removes the filter → the system restores all offerings of the semester.

A4 (step 4): Meeting time unrecorded → the system shows “not available”.

Exception Flows:

E1 (step 3): Data retrieval fails → the system displays an error notice (no stale data) and allows a retry.

E2 (step 5): Invalid filter value → the system displays a validation message.

E3 (step 2): No semester selected → the system asks the student to select a semester.

Postconditions: The student has viewed the course offerings. No application data is created or modified.

UC-02 — Check Course Prerequisites

Actor: Student

Related Requirement: FR-02

Goal: Check whether the student’s completed courses meet the prerequisites of a selected course.

Preconditions: The student is authenticated and the selected target course exists.

Trigger: The student requests a prerequisite check, or UC-03 invokes it automatically during schedule building.

Main Success Flow: 1. The student chooses a course and asks whether it has prerequisites. 2. The system fetches the course’s prerequisite rules and the student’s course records. 3. The system matches the requirements against the history. 4. The system displays the per-prerequisite breakdown, any data notice, and the result (“satisfied”).

Alternative Flows: A1 (step 2): The target course has no prerequisites → “satisfied” (none required). A2 (step 4): One or more prerequisites are not met → the system shows “not satisfied” and lists the unmet prerequisites. A3 (steps 2–3): The history is missing or the prerequisite rule is not defined → the system shows “cannot be determined”.

Exception Flows: E1 (steps 2–3): Data retrieval fails → the system shows an error message without an evaluation result.

Postconditions: The student has been evaluated as “satisfied”, “not satisfied”, or “cannot be determined”. No records are modified.

UC-03 — Create Semester Schedule

Actor: Student

Related Requirement: FR-03

Includes: UC-02 (Check Prerequisites), UC-04 (Detect Conflicts)

Goal: Build a proposed semester schedule, edit and save it, with live feedback on prerequisites, conflicts, capacity, and data warnings.

Preconditions: The student is authenticated and the sections of the planning semester are active.

Trigger: The student starts a new schedule or opens an existing saved schedule.

Main Success Flow: 1. The system shows the sections available for the selected semester. 2. The student selects a section to add, and it is added to the workspace grid. 3. The system runs UC-02 for the course of each chosen section («include»). 4. The system runs UC-04 on all selected sections («include»). 5. The system checks section capacity and status. 6. The system renders the workspace grid, itemized warnings, the validation status ("No issues detected" / "Has warnings"), the planning notice, and the data notice. 7. The student adds or removes sections; each change re-runs the prerequisite and conflict checks. 8. The student submits a save request; the system saves the proposed schedule (FR-11) and displays a confirmation.

Alternative Flows: A1 (step 4): Selected sections overlap → the system shows a conflict warning with an explanation of the overlap. A2 (step 3): Prerequisite not met or undetermined → a prerequisite warning is shown. A3 (step 5): Section full → a full-section warning is shown. A4 (steps 3–5): Data missing → an incomplete-data warning is shown and the status is never "No issues detected". A5 (step 1): The student selects "Open a saved plan" → the system loads it and re-runs the checks (FR-12). A6 (step 8): If the team allows saving plans with warnings, the plan is saved with a warning state (subject to team decision).

Exception Flows: E1 (step 8): No sections selected or save failure → the system prompts for a selection or shows a save error without marking the plan as saved (FR-11). E2 (steps 1–2): Data retrieval fails → an error notice is shown.

Postconditions: A proposed schedule is saved. Its status is re-validated when it is loaded or rechecked under FR-12, and is not assumed to be remembered unless FR-11 stores it. No official registration takes place; course offerings, seats, and history do not change.

UC-04 — Detect Schedule Conflicts

Actor: Student

Related Requirement: FR-04

Goal: Identify overlapping weekly class meeting times among the sections selected in a proposed schedule.

Preconditions: The proposed-schedule workspace is open and the application is available.

Trigger: The student requests a conflict check, or UC-03 invokes this use case automatically during schedule building.

Main Success Flow: 1. The system retrieves the meeting days and times of the selected sections. 2. The system validates the meeting times (day; start < end). 3. The system compares the meeting times of all section pairs on the same day. 4. The system calculates the overlapping pairs and overlap intervals. 5. The system displays the detected conflicts (sections, day, overlap interval) or confirms "No time conflicts detected".

Alternative Flows: A1 (step 3): Sections meet back-to-back (end1 = start2) → no conflict is reported. A2 (step 2): A meeting time is missing or start ≥ end → the system reports "Incomplete check" naming the affected sections, while still reporting valid conflicts. A3 (step 1): Fewer than two sections are selected → the system reports "nothing to compare".

Exception Flows: E1 (step 1): Section data retrieval fails → the system displays an error notice.

Postconditions: The student receives a conflict detection report. In UC-03, the output is integrated into the workspace warning panel.

UC-05 — Calculate GPA

Actor: Student

Related Requirement: FR-05

Input: The recorded completed courses, their credit hours, and the project grade scale.

Preconditions: The student is authenticated; the grade-point scale, eligibility policy, and rounding rule are defined.

Trigger: The student requests a GPA calculation.

Main Success Flow: 1. The system retrieves the student's completed-course records. 2. The system filters the courses according to the policy. 3. The system checks the grades and credit hours of the records. 4. The system calculates the weighted GPA and rounds it using the rounding rule. 5. The system shows the estimated GPA, the course and credit counts, the list of excluded courses, and the estimate notice.

Alternative Flows: A1 (step 2): Non-eligible courses appear in the history → the system lists the excluded courses with the reason and calculates the GPA from the eligible courses.

Exception Flows: E1 (step 3): A grade or credit-hours value is invalid or missing → the system reports the failing record and displays no GPA. E2 (step 2): No eligible courses, or total credits = 0 → the system displays an explanatory message and no GPA. E3 (step 1): System or data retrieval failure → the system shows an error message without a GPA.

Postconditions: The student has an estimated GPA with a calculation summary. Course records and grades are not altered.

UC-06 — View Course/Section Details

Actor: Student

Related Requirement: FR-06

Extends: UC-01 (Display Course Offerings)

Goal: Inspect the recorded details of a course or section without modifying the proposed plan.

Preconditions: The student is viewing the course and section results (UC-01), and the selected record has an identifier supplied by the results list.

Trigger: The student selects a course or section from the course offering results.

Main Success Flow: 1. The student selects a course or section. 2. The system validates the selected identifier. 3. The system retrieves the current stored record. 4. The system displays the recorded course code, name, credit hours, section identifier, meeting days and times, capacity and status, and prerequisite information where available. 5. The system marks missing fields as "not available"; missing prerequisite data is not represented as "no prerequisites". 6. The student reviews the details and may return to the results.

Alternative Flows: A1 (step 5): A field is not recorded → the system marks each unavailable field and continues to display the other recorded details.

Exception Flows: E1 (step 2): The identifier is unknown or stale → the system displays an error and returns the student to the results. E2 (step 3): The record cannot be retrieved → the system reports that the details could not be retrieved and does not show cached details as current.

Postconditions: The selected record's recorded details have been displayed; the proposed plan and the source records are unchanged. After an error, an explanation is displayed and no stale record is presented as current.

Dependencies: FR-01 supplies the result list; supporting system: the course and section data store. GR-1 (missing data is not inferred) and GR-3 (read-only inspection) apply.

UC-07 — Update Proposed Plan Selection

Actor: Student

Related Requirement: FR-07

Extends: UC-03 (Create Semester Schedule)

Goal: Add or remove a section in a proposed semester plan without creating duplicate selections or changing official university records.

Preconditions: A proposed plan and its semester are selected, the target section identifier is available, and the plan is a proposed application record, not an official registration.

Trigger: The student chooses Add or Remove for a section while viewing a proposed plan.

Main Success Flow: 1. The student chooses Add or Remove for a section. 2. The system verifies that the section belongs to the selected semester and is available in the application's recorded data. 3. For Add, the system checks whether the same section is already selected. 4. If the section is valid and not a duplicate, the system adds it; for Remove, the system removes the section from the proposed plan. 5. The system invokes the validation workflow of FR-03, including the checks of FR-02 and FR-04 and the section-status and data-completeness checks. 6. The system displays the updated proposed plan, a confirmation, and the refreshed warnings and status.

Alternative Flows: A1 (step 2): The section belongs to another semester or is unavailable → the system rejects the change and explains the reason. A2 (step 3): The section is already selected → the system does not add a second copy and explains that it is already selected. A3 (step 4): Removal → the system removes only the selected section and recalculates the affected warnings and totals. A4 (step 5): Validation data is incomplete → the system identifies the affected item and shows an incomplete or warning status, never "No issues detected".

Exception Flows: E1 (step 4): The update fails → the system reports the failure, leaves the selection unchanged, and shows no success confirmation.

Postconditions: The proposed selection is updated and the warnings and validation status are refreshed. After a failure, the selection is unchanged and the reason is explained.

Dependencies: Supporting services: the proposed-plan and course/section data services. Coordinates with FR-03 and does not define a second schedule-saving workflow. Whether a plan with warnings may be saved remains pending team agreement (FR-03).

UC-08 — Review Plan and Credit-Hour Total

Actor: Student

Related Requirement: FR-08

Extends: UC-03 (Create Semester Schedule)

Goal: View the selected sections and a correct credit-hour total for the current proposed plan.

Preconditions: A proposed plan is selected, and the selected section and course records and their credit-hour values can be retrieved, or their absence can be reported.

Trigger: The student opens the plan summary, or a selection change refreshes the summary.

Main Success Flow: 1. The student opens the proposed-plan summary. 2. The system retrieves the selected sections and their recorded credit-hour values. 3. The system verifies that each required credit-hour value is valid and that no selected item is double-counted. 4. The system sums the credit hours of the distinct selected courses according to the agreed course and section data model. 5. The system displays the selected sections, their recorded meeting details, the total credit hours, and any relevant incomplete-data notice.

Alternative Flows: A1 (step 5): The plan is empty → the system displays the empty state and zero selected sections, and displays a zero total only if the agreed display policy permits. A2 (step 3): A credit-hour value is missing or invalid → the system identifies the affected course or section and marks the total incomplete; it does not count the value as zero.

Exception Flows: E1 (step 2): The data cannot be retrieved → the system displays an error and does not present a stale total as current.

Postconditions: The selected sections and the computed total, or an incomplete status, are displayed. No course, section, or official record is changed by viewing the summary.

Dependencies: Supporting services: the proposed-plan and course/section data services. The credit-hour total is not a GPA; FR-05 remains responsible for GPA. The team must confirm whether credit hours belong to course or section records.

UC-09 — Review Conflict Results

Actor: Student

Related Requirement: FR-09

Includes: UC-04 (Detect Schedule Conflicts)

Goal: View the results of the conflict check of UC-04 in a way that lets the student identify each conflict.

Preconditions: A proposed plan is open, and the system can access the current plan and its meeting-time data, or can identify missing or invalid data.

Trigger: The student requests conflict results, or the proposed plan changes and the validation workflow of UC-03 refreshes.

Main Success Flow: 1. The system obtains the latest conflict-check result of FR-04 for the current proposed plan. 2. The system lists every reported conflicting pair. 3. For each pair, the system displays both course and section identifiers, the day, and the overlapping interval. 4. If the check completed and found no conflicts, the system displays the no-conflict result. 5. The student reviews the results; viewing them does not change the plan.

Alternative Flows: A1 (step 3): Meeting data is missing or invalid → the system shows an incomplete-check warning and identifies the affected sections. A2 (step 2): Fewer than two sections are selected → the system indicates that there are no pairs to compare and does not imply that unavailable data was checked successfully. A3 (step 1): A section is added or removed → the system refreshes the results using the current plan and does not leave earlier results marked as current.

Exception Flows: None beyond those inherited from UC-04.

Postconditions: Current conflict results, or a valid no-conflict result, are displayed. If the check is incomplete, the system says so and does not claim that the plan is conflict-free.

Dependencies: FR-04 remains authoritative. Conflict rule: same day and start1 < end2 and start2 < end1; back-to-back intervals do not conflict. This use case presents results and does not define a second conflict algorithm.

UC-10 — Review Validation Summary

Actor: Student

Related Requirement: FR-10

Extends: UC-03 (Create Semester Schedule)

Goal: View one current summary of the prerequisite, conflict, section-status, and incomplete-data findings for the proposed plan.

Preconditions: A proposed plan is selected, and the validation checks can be run, or their incomplete or failed state can be identified.

Trigger: The student reviews the plan, or a plan change refreshes the validation.

Main Success Flow: 1. The system collects the latest available prerequisite, conflict, section-status, and data-completeness results for the current plan. 2. The system associates each finding with the affected course or section. 3. If all required checks complete and no issue is found, the system displays "No issues detected". 4. If any issue or incomplete check exists, the system displays "Has warnings" and lists the findings and notices. 5. The system keeps the planning notice visible so that the student does not mistake the plan for an official registration.

Alternative Flows: A1 (step 1): A check cannot complete → the system identifies the affected item and displays a warning; unknown data is not treated as success. A2 (step 1): The student changes the plan → the system recalculates the summary and removes only the warnings that the current results no longer support.

Exception Flows: E1 (step 1): The validation results cannot be retrieved → the system displays an error and does not present earlier results as current.

Postconditions: A current summary is shown as "No issues detected" or "Has warnings", with the relevant details and notices. A failed check is never represented as successful.

Dependencies: Supporting use cases and services: UC-03, FR-02, and FR-04 validation outputs. Coordinates with FR-03 and GR-1 to GR-3; it summarizes the existing rules and does not replace or contradict them.

UC-11 — Save Proposed Schedule

Primary Actor: Student

Goal: Persist the active schedule to the student’s profile.

Related Requirement: FR-11

Preconditions: The student is authenticated and has an active schedule view open.

Main Success Flow: 1. The student selects “Save Schedule”. 2. The system checks that the workspace contains sections. 3. The system saves the section IDs and displays a confirmation.

Alternative Flow: The workspace is empty → the system prompts the student to select sections first.

Exception Flow: A database error occurs → the system displays a failure message.

UC-12 — Load Saved Schedule

Primary Actor: Student

Goal: Display a stored semester schedule.

Related Requirement: FR-12

Preconditions: The student is authenticated and has existing schedule data.

Main Success Flow: 1. The student selects a semester. 2. The system retrieves the stored section IDs. 3. The system re-evaluates conflicts and prerequisites, then renders the workspace grid.

Alternative Flow: No saved plan exists → the system displays an empty workspace.

Exception Flow: A retrieval error occurs → the system displays an error message.

UC-13 — Clear Schedule Plan

Primary Actor: Student

Goal: Empty the workspace grid.

Related Requirement: FR-13

Preconditions: The workspace contains at least one section.

Main Success Flow: 1. The student selects “Clear Schedule”. 2. The system requests confirmation. 3. The student confirms; the system clears the grid and resets the metrics.

Alternative Flow: The student cancels the confirmation → the system leaves the workspace intact.

UC-14 — Export Schedule Plan

Primary Actor: Student

Goal: Download an offline summary PDF of the current plan.

Related Requirement: FR-14

Preconditions: The workspace contains at least one section.

Main Success Flow: 1. The student selects “Export Schedule”. 2. The system builds a formatted PDF document. 3. The system triggers the file download in the browser.

Alternative Flow: The workspace is empty → the export option is disabled.

Exception Flow: PDF rendering fails → the system alerts the user.

UC-15 — Reset Student History

Primary Actor: Student

Goal: Clear the recorded completed courses and grades.

Related Requirement: FR-15

Preconditions: Completed courses exist in the history records.

Main Success Flow: 1. The student selects “Reset History”. 2. The system requests confirmation. 3. The student confirms; the system deletes the entries and updates the GPA display.

Alternative Flow: The student cancels the confirmation → the system aborts the request.

UC-16 — View Degree Progress & Run Degree Audit

Actors: Primary: Student. Secondary: UIS.

Description: The student sees the completed, in-progress, and missing requirements, along with the courses that qualify for those requirements.

Preconditions: The student is authenticated; academic data (or its synchronization age) is available.

Normal Flow: 1. The student clicks Degree Progress. 2. The system loads the program, requirements, and enrollments. 3. The system assigns courses to requirements and computes the credits and progress. 4. The system displays the remaining courses that qualify. 5. The student optionally exports the audit to PDF.

Alternative / Exception Flows: 2a. No program found → the system displays the advisor's contact message. 2b. UIS unavailable or data stale (> 24 h) → the system displays the data date and uses the available data.

Relationships: «include» Validate Prerequisites (FR-02); «include» Export Audit to PDF.

Postconditions: The audit and progress report is displayed; a PDF of the report is created when requested.

UC-17 — Request and Process an Advisor Override

Actors: Primary: Student and Academic Advisor. Secondary: Notification Service, Audit Log.

Description: The student requests an exception, and the advisor decides on it.

Normal Flow: 1. The student selects the section and the override type. 2. The student writes a justification and submits. 3. The system receives the request and routes it to the advisor. 4. The advisor reviews the request. 5. The advisor approves or declines it, or asks for more information, adding comments. 6. The system stores and logs the decision and notifies the student.

Alternative / Exception Flows: 2a. No justification, or an invalid section → the submission is not accepted. 3a. An active request already exists for the same section → the request is blocked. 5a. The advisor does not respond or asks for more information → the request remains pending.

Relationships: «include» Send Event Notification (UC-18).

Postconditions: The decision is saved and the student is notified. An approved override applies only to that student and section.

UC-18 — Send Event Notification

Actors: Primary: System event service. Secondary: Recipient user.

Description: The system informs a user of a business event through in-app and/or email channels.

Preconditions: A recognized event occurs; the recipient and channel policy can be determined.

Normal Flow: 1. An event is raised. 2. The system maps it to a template and recipient. 3. The system checks the recipient's channel preferences. 4. The system sends the in-app notice and/or email. 5. The system records the notification.

Alternative / Exception Flows: 2a. The recipient is invalid → the failure is recorded. 3a. Email is disabled by the user → the email is suppressed. 4a. The email gateway fails → the system retries, then marks the notification FAILED and logs it.

Relationships: Invoked by UC-17 and other registration use cases.

Postconditions: The notification status, timestamp, and retry count are stored.

UC-19 — Manage Roles and Permissions

Actors: Primary: System Administrator. Secondary: Authorization Service, Audit Log.

Description: The administrator assigns permissions (resource, action, scope) to roles.

Preconditions: The administrator is logged in with administrative rights.

Normal Flow: 1. The administrator opens access control. 2. The administrator chooses a role, resource, action, and scope. 3. The system validates the values. 4. The administrator saves. 5. The system applies the mapping and logs the change.

Alternative / Exception Flows: 3a. Invalid mapping → the system displays a validation message. 1a. A non-administrator user who tries to access this function is denied with HTTP 403.

Relationships: «include» Write Audit Log.

Postconditions: The role–permission mappings are updated and audited.

UC-20 — Configure Academic Term & Registration Windows

Actors: Primary: Academic Administrator / Department Chair. Secondary: Registration Service, Audit Log. Description: The actor sets up a term and its registration windows, which determine who can register and when. Preconditions: The actor is authenticated with administrative rights.

Normal Flow: 1. The actor opens term management. 2. The actor enters the term dates and the add/drop deadline. 3. The actor defines the registration window dates and credit ranges. 4. The system validates the values. 5. The actor saves and publishes. 6. The scheduler opens and closes the windows and checks registration eligibility.

Alternative / Exception Flows: 4a. Invalid dates, ranges, or overlaps → the system rejects them and gives the reason. 6a. A student outside the window or range tries to register → registration is denied. 6b. A scheduler error occurs : the error is logged and the system fails safely.

Relationships: «include» Write Audit Log.

Postconditions: The term and windows are saved with their statuses and are enforced at registration.

3.3 Conceptual Class Model

3.3.1 Conceptual Class Diagram

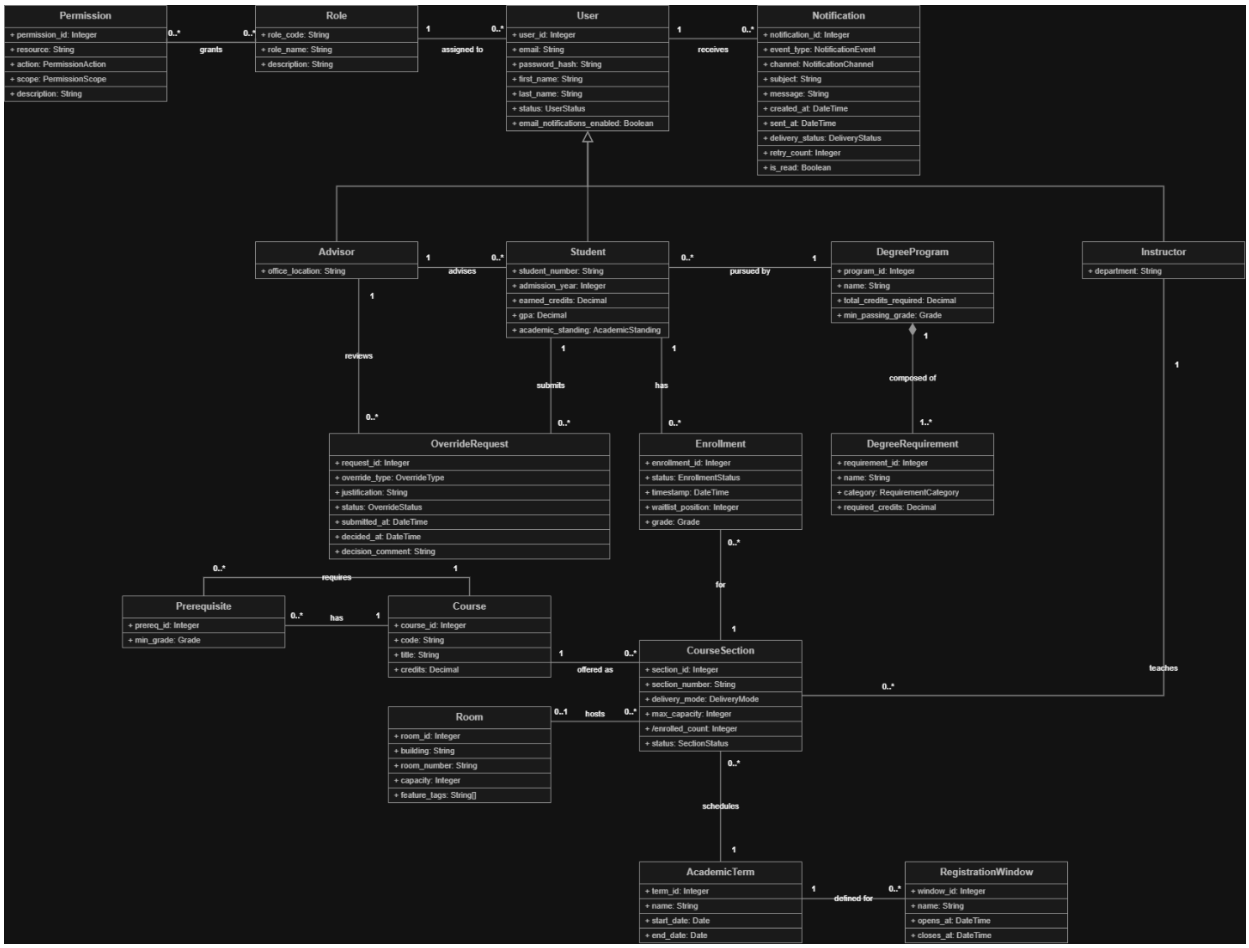


Figure 4: Conceptual class diagram

3.3.2 Class Descriptions

Class	Attributes	Relationships
User		Base class that specializes into Student and Advisor; stores the last login time. Associations: Role 1–0..* User; Department 0..1–0..* User; User 1–0..* Notification; User 0..1–0..* AuditLog; User 0..1–0..* CourseSection (teaches).
Role	role_code, role_name, description	Named access role. Role 0..*–0..* Permission (grants); every User has one Role.
Permission	permission_id, resource, action, scope, description	The right a role must hold to perform an action on a resource within a scope.
Student	student_number, admission_year, earned_credits, gpa, academic_standing	Academic profile. Student 0..*–0..1 DegreeProgram; 0..*–0..1 Advisor; 1–0..* Enrollment, DraftSchedule, and OverrideRequest.
Advisor	office_location	Advises students and grants or denies their requests. Advisor 1–0..* OverrideRequest (reviews).

Class	Attributes	Relationships
Department	department_id, code, name	Organizational unit. 1–0..* DegreeProgram (offers); 1–0..* Course (owns).
DegreeProgram		Degree definition. Composition: 1–1..* DegreeRequirement.
DegreeRequirement		A credit group of a program. 0..*–1..* Course (satisfied by).
Course	course_id, code, title, credit_hours, description	Catalog subject. 1–0..* CourseSection (offered as); linked to itself through Prerequisite (a course that must be completed before another; 0..* in each direction).
AcademicTerm	term_id, name, start_date, end_date, add_drop_deadline, status	Teaching period. 1–0..* CourseSection; 1–0..* RegistrationWindow; 1–0..* DraftSchedule.
RegistrationWindow	registration_window_id, registration_window_name, min_earned_credits, max_earned_credits, open_at, close_at, status	Specifies when students within a credit range may register; belongs to one AcademicTerm.
CourseSection	section_id, section_number, delivery_mode, max_capacity, enrolled_count, status	The offering of a Course in a Term. 0..*–0..1 Room; composition 1–0..* MeetingTime; 1–0..* Enrollment and OverrideRequest.
MeetingTime	meeting_id, day_of_week, start_time, end_time	A weekly time slot of a CourseSection.
Room	room_id, building, room_number, capacity, feature_tags	Teaching space that hosts 0..* CourseSections.
Enrollment	enrollment_id, status, timestamp, waitlist_position, grade	Links one CourseSection to one Student and tracks whether the student is enrolled, waitlisted, or dropped.
DraftSchedule	draft_id, name, created_at, updated_at	A plan saved by a Student for one Term, containing 0..* CourseSections (many-to-many).
OverrideRequest	request_id, override_type, justification, status, submitted_at, decided_at, decision_comment	Submitted by one Student for one CourseSection and reviewed by one Advisor.
Notification	notification_id, event_type, channel, subject, message, created_at, sent_at, delivery_status, retry_count, is_read	An in-app or email message addressed to a single User.
AuditLog	log_id, event_type, entity_type, entity_id, ip_address, occurred_at, payload	Immutable record of system actions; created by 0..1 User.

3.4 Non-Functional Requirements

NFR-01 — Course Search Performance

Requirement Statement: For the specified test dataset and environment, at least 95% of valid course-search requests shall return a visible result within 2.0 seconds.

Type & Status: Performance / Efficiency requirement [2]. Target: ≤ 2.0 seconds at a success rate of ≥ 95%. Test status: not yet tested.

Rationale & Scope: Students need fast feedback when planning schedules, as they search repeatedly. Applies to valid FR-01 searches with a semester selection and an optional code/name filter.

Test Environment & Dataset: Documented before the test: server hardware/hosting, OS, database, browser version, client device, and network configuration. The dataset contains a realistic set of courses and sections agreed by the team. All 100 requests use the same dataset and environment.

Test Method & Measurement: (1) Run 100 search queries (code filter, name filter, combinations, with and without results). (2) Execute them sequentially after a single warm-up, with no concurrent load. (3) Use browser developer tools or a timing script to measure from the moment the search is submitted until the correct result or the "no match" message appears. Record the tool and the resolution used (milliseconds or, if less precise, 0.1 second). An individual request passes if it completes in ≤ 2.0 seconds and shows accurate results or the unrecorded / no-match message.

Pass / Fail Decision: PASS when ≥ 95 of 100 requests pass; FAIL if ≤ 94 pass. Reported as "Not Tested" until implemented.

Consistency: NFR-01 measures search response time, while NFR-04 measures schedule-export time. The requirements refer to different operations and do not conflict.

NFR-02 — Security

Priority: Essential

Unauthenticated access to protected resources shall return HTTP 401. A request from a user who lacks the required role or scope shall be denied with HTTP 403 and without data leakage. No High or Critical vulnerability (CVSS ≥ 7.0) shall remain open at release, and all traffic shall use TLS 1.3.

Tests and Success Criteria: (1) RBAC matrix: deploy to staging with Student, Advisor, Instructor, Chair, and Admin accounts and automatically check 401/403/allow outcomes for every account and scope; success = 100% of checks correct, no protected data in any denied response, and every denial logged. (2) Transport: scan HTTPS/WSS with testssl.sh, including TLS 1.2 attempts and a plain HTTP request; success = only TLS 1.3 is accepted. (3) Vulnerability scan: scan the release build (authenticated and unauthenticated); success = no High/Critical (CVSS ≥ 7.0) findings.

NFR-03 — Availability & Recoverability

Priority: High

Monthly availability shall be at least 99.9% (no more than 43.2 minutes of unplanned downtime in a 30-day period), and data loss shall not exceed 60 seconds of acknowledged transactions per month.

Tests and Success Criteria: (1) Soak: a 30-day availability test on staging with ≥ 3 replicas, a PostgreSQL primary/replica, and WAL archiving ≤ 60 s; success = availability ≥ 99.9%. (2) Fault injection: kill the application pod, the PostgreSQL primary, and a Redis node individually (at least 10 runs each); success = service restored within 5 minutes in 100% of runs. (3) Integrity and restore:

data loss ≤ 60 s; reconcile the loss against the count of ENROLLED transactions and run a point-in-time restore; success = no count mismatches and the integrity checks pass.

NFR-04 — Schedule Export Performance

Type: Performance / Efficiency requirement

Requirement: Under the documented test conditions, at least 95 out of 100 valid PDF export requests (with up to 6 sections) shall trigger the file download within 1.5 seconds.

Test Method: Execute 100 sequential automated export requests on standard client setups.

NFR-05 — Schedule Validation Response Time

Requirement Statement: Under a documented test environment, at least 95 out of 100 valid validation requests for a proposed schedule containing up to 50 section meeting entries shall display the complete validation result within 2.0 seconds.

Type & Status: Performance / Efficiency requirement [2]. Proposed project target, pending team agreement; the values are team proposals, not values mandated by the project handout. Test status: not yet tested.

Rationale & Scope: Students need prompt feedback while adjusting schedules. Applies to plan validation (FR-03 and FR-10), measured from submitting the validation until the result and status are visible. This requirement concerns plan validation, not course search (NFR-01).

Test Environment & Dataset: Before the test, the report states: server hardware or hosting, operating system, database, browser and version, client device, and network type. The report also states the dataset: up to 50 meeting entries, including overlaps, non-overlaps, different days, adjacent times, and missing or invalid times. The same environment and dataset are used for all 100 requests.

Test Method & Measurement: (1) Use one fixed environment and the documented dataset. (2) Perform one unmeasured warm-up request, then 100 sequential valid requests from one client. (3) Record the elapsed time of each request in milliseconds. (4) Record the timing tool, and the expected and actual result of each request. A request is successful if the correct complete result appears in 2.0 seconds or less.

Pass / Fail Decision: PASS when at least 95 of the 100 requests are successful; FAIL if 94 or fewer are successful. Reported as "Not Tested" until the test has been run; the requirement must not be reported as passed until recorded evidence shows a pass.

Consistency: NFR-01 (course search), NFR-04 (schedule export), and NFR-05 (plan validation) use proposed values for different operations. Any further performance requirement must be checked against all three to avoid conflicting thresholds.

3.5 External Interface Requirements

3.5.1 User Interfaces

The web layout features a main header navigation bar, an interactive grid workspace displaying weekly class schedules, a sidebar showing summary statistics and validation warnings, and standard input controls such as search inputs, dropdown filters, and buttons.

3.5.2 Hardware Interfaces

Client: Standard desktop or mobile computing devices equipped with displays, keyboards, and pointing tools.

Server: Web and database hosting infrastructure capable of serving concurrent HTTPS traffic.

3.5.3 Software Interfaces

Web client and application server: Communicate via RESTful APIs returning standard JSON payloads.

Database: The server connects to PostgreSQL using standard relational database drivers.

3.5.4 Communications Interfaces

Encrypted web protocol (HTTPS) running over standard TCP/IP port 443.

3.6 Traceability

The following entries identify the related use cases and acceptance or test coverage for FR-06 to FR-10 and NFR-05.

1. FR-06 — Use case: UC-06, which extends UC-01. Acceptance coverage: AC-06.1 to AC-06.3, including correct record display, missing values, and read-only behavior.

2. FR-07 — Use case: UC-07, which extends UC-03. Acceptance coverage: AC-07.1 to AC-07.4, including adding and removing sections, duplicate prevention, and no change to official records.

3. FR-08 — Use case: UC-08, which extends UC-03. Acceptance coverage: AC-08.1 to AC-08.3, including the credit-hour total, removal of a section, and invalid data.

4. FR-09 — Use case: UC-09, which includes UC-04. Acceptance coverage: AC-09.1 to AC-09.4, including overlapping intervals, back-to-back meetings and different days, incomplete data, and refreshed results.

5. FR-10 — Use case: UC-10, which extends UC-03. Acceptance coverage: AC-10.1 to AC-10.4, including clean status, warnings, refreshed status, and notices.

6. NFR-05 — Related use cases: UC-09 and UC-10 (validation). Test coverage: 100 sequential runs; the requirement passes when at least 95 requests return correct results within 2.0 seconds.

4. Appendices

4.1 Appendix A: Data Dictionary (Table Names)

Each table contains a primary key; foreign keys follow the associations in Section 3.3.1.

Table	Purpose
Roles	Named access roles
Permissions	Resource/action/scope rights
RolePermissions	Role–permission mapping
Users	Accounts and notification preference
Students	Student academic profile
Advisors	Advisor profile
Departments	Academic departments
DegreePrograms	Degrees and credit totals
DegreeRequirements	Credit groups of a program
RequirementCourses	Courses satisfying a requirement
Courses	Course catalog
Prerequisites	Course dependency rules
AcademicTerms	Terms and deadlines
RegistrationWindows	Registration periods and credit ranges
Rooms	Rooms, capacity, features
CourseSections	Term offerings of courses
MeetingTimes	Weekly section time slots
Enrollments	Student–section status, waitlist, grade
DraftSchedules	Saved student plans
DraftScheduleItems	Sections inside a draft
OverrideRequests	Override requests and decisions
Notifications	Messages and delivery state
AuditLogs	Append-only event history

4.2 Appendix B: Glossary

Term	Definition
Academic Standing	Status of a student's standing (good standing, probation, suspension).
Advisor Override	An exception to a prerequisite or load rule, approved by the advisor.
Audit Log	Unmodifiable, time-stamped record of actions in the system.
Corequisite	A course that may be taken in the same term as the dependent course.

Term	Definition
CVSS	Common Vulnerability Scoring System; rates vulnerability severity from 0 to 10.
Degree Audit	Comparison of a student's courses against the program requirements.
Draft Schedule	A saved "what-if" selection of sections.
RBAC	Role-Based Access Control: permissions are assigned to roles.
Registration Window	The period during which a student group may register.
RPO / RTO	Recovery point objective (maximum data loss) / recovery time objective (maximum restore time).
TLS 1.3	Protocol version that encrypts network traffic.
WSS	WebSocket Secure: an encrypted real-time channel.